



k8s Workshop

Part 3 ConfigMaps & Secrets

Advantages of running apps in containers

- reducing the gap between environments
- the same image is used from DEV -> PRD
- each env uses the same set of binaries
- differences between environments is left configuration

Configuring APP in k8s

- ConfigMap
- Secrets

ConfigMap

- Environment Variables
- Text Files (json, yaml, ...)

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: hello-world-config-env-vars
data:
  K8S_WS_PART: part-03
  K8S_WS_MODE: awesome
```

```
kind: Deployment
spec:
  ...
  containers:
  - name: app-hello-world
    ...
    envFrom:
    - configMapRef:
        name: hello-world-config-env-vars
```

Text Files

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: hello-world-config-files
data:
  hello-world.yaml: |
    app:
      name: hello-world
      version: 1.0.0
  hello-world-2.yaml: |
    app:
      name: hello-world-2
      version: 2.0.0
```

```
apiVersion: apps/v1
kind: Deployment
spec:
  ...
  containers:
  - name: app-hello-world
    ...
    volumeMounts:
    - name: config-volume
      mountPath: /app/config
  volumes:
  - name: config-volume
    configMap:
      name: hello-world-config-files
```

Secrets

- Same API as ConfigMap
- Only sent to nodes that need to use them
- Encrypted both in transit and at rest
- People with access to the resources can see the value
- Obfuscated with base64 encoding

Secrets format

- Environment Variables
- Text Files (json, yaml, ...)

```
apiVersion: v1
kind: Secret
metadata:
  name: hello-world-secret-vars
type: Opaque
stringData:
  K8S_WS_PASSWORD: mypassword
```

```
apiVersion: apps/v1
kind: Deployment
spec:
  ...
  containers:
    - name: app-hello-world
      secretRef:
        name: hello-world-secret-vars
```

```
apiVersion: v1
kind: Secret
metadata:
  name: hello-world-secret-files
type: Opaque
stringData:
  secret.yaml: |
    app:
      password: mypassword
  secret2.yaml: |
    app:
      password: mypassword-2
```

```
apiVersion: apps/v1
kind: Deployment
spec:
  ...
  containers:
  - name: app-hello-world
    volumeMounts:
    - name: secret-volume
      mountPath: /app/secret
  volumes:
  - name: secret-volume
    secret:
      secretName: hello-world-secret-files
```

Secrets and VCS (git)

- secrets should not contain actual secrets but placeholders
- replaceable from pipeline (AKV + ReplaceSecrets task)